



UNIVERSITY OF NAIROBI

UNIVERSITY OF NAIROBI

DEPARTMENT OF MECHANICAL AND MANUFACTURING ENGINEERING

Analytical Hand Calculations & Theoretical Predictions

Project: Bridging the Gap: Computational Prediction vs. Physical Performance of 3D-Printed Lattice Structures

Student: Geoffrey Mulu Mutuku

Registration Number: F18/144332/2022

1. Fundamental Assumptions

To perform analytical predictions based on the Gibson-Ashby model for cellular solids, the following material properties for Polylactic Acid (PLA) and geometric assumptions are established:

- **Solid Density** (ρ_s): $1.24 \text{ g/cm}^3 = 0.00124 \text{ g/mm}^3$
- **Solid Young's Modulus** (E_s): 2500 MPa
- **Solid Yield Strength** (σ_{ys}): 40 MPa
- **Topology**: Octet-Truss (Stretching-dominated behavior)
- **Bounding Envelope**: $60 \text{ mm} \times 60 \text{ mm} \times 60 \text{ mm}$ (Core Lattice)
- **Top and Bottom Caps**: $60 \text{ mm} \times 60 \text{ mm} \times 2 \text{ mm}$ (each)
- **Unit Cell Size** (L): 20 mm
- **Assumed Strut Diameter** (d): 2.0 mm (Radius $r = 1.0 \text{ mm}$)
- **Node Overlap**: Ignored for simplified theoretical volume derivations.

2. Geometric Calculations

2.1. Strut Length

For an octet-truss unit cell, the struts form face-centered diagonal elements. The length of a single strut (l) relates to the cubic unit cell size (L) by:

$$l = \frac{L}{\sqrt{2}}$$
$$l = \frac{20}{\sqrt{2}} = 14.14 \text{ mm}$$

2.2. Analytical Relative Density

The relative density ($\bar{\rho}$) of a lattice is the ratio of the lattice density to the solid material density (ρ^*/ρ_s). For an idealized octet-truss lattice, it is derived mathematically as:

$$\bar{\rho} = 6\sqrt{2}\pi \left(\frac{r}{l}\right)^2$$
$$\bar{\rho} = 6 \times 1.414 \times 3.1416 \times \left(\frac{1.0}{14.14}\right)^2$$
$$\bar{\rho} = 26.65 \times 0.0050 = 0.133$$

The lattice operates at a relative density of approximately 13.3%.

3. Mass and Volume Predictions

3.1. Total Envelope Volume

Lattice Core Volume (V_{core}):

$$V_{core} = 60 \times 60 \times 60 = 216,000 \text{ mm}^3$$

Solid Caps Volume (V_{caps}):

$$V_{caps} = 60 \times 60 \times 4 \text{ (total thickness)} = 14,400 \text{ mm}^3$$

3.2. Predicted Prototype Mass

Lattice Core Mass (M_{core}):

$$M_{core} = V_{core} \times \bar{\rho} \times \rho_s$$
$$M_{core} = 216,000 \times 0.133 \times 0.00124 = 35.62 \text{ g}$$

Caps Mass (M_{caps}):

$$M_{caps} = 14,400 \times 0.00124 = 17.86 \text{ g}$$

Total Predicted Mass:

$$M_{total} = 35.62 + 17.86 = 53.48 \text{ g}$$

4. Mechanical Behavior Predictions

The octet-truss is a stretching-dominated structure. According to the Deshpande and Fleck modification of the Gibson-Ashby model for octet lattices, the effective mechanical properties scale linearly with relative density.

4.1. Effective Young's Modulus

The analytical scaling constant for an octet truss is approximately $C_1 = 0.11$.

$$E^* = C_1 E_s \bar{\rho}$$

$$E^* = 0.11 \times 2500 \text{ MPa} \times 0.133$$

$$E^* = 36.58 \text{ MPa}$$

4.2. Theoretical Global Stiffness

Using Hooke's Law for a macroscopic block ($A = 60 \times 60 = 3600 \text{ mm}^2$, initial height $H = 60 \text{ mm}$ core):

$$k = \frac{E^* A}{H}$$

$$k = \frac{36.58 \times 3600}{60}$$

$$k = 2194.8 \text{ N/mm}$$

Theoretical prediction indicates the specimen will compress by 1 mm for every 2.19 kN of applied force.

4.3. Effective Yield Strength

Using the scaling constant $C_2 \approx 0.3$ for stretching-dominated lattice yield:

$$\sigma_y^* = C_2 \sigma_{ys} \bar{\rho}$$

$$\sigma_y^* = 0.3 \times 40 \text{ MPa} \times 0.133$$

$$\sigma_y^* = 1.6 \text{ MPa}$$

Predicted Global Yield Load (F_y):

$$F_y = \sigma_y^* \times A$$

$$F_y = 1.6 \times 3600 = 5760 \text{ N}$$

5. Micro-Structural Stability (Strut Buckling)

To determine if individual struts will buckle before they yield in compression, Euler's critical buckling load (P_{cr}) is calculated for a single strut. Assuming fixed-fixed boundary conditions at the nodes ($K = 0.5$):

Area Moment of Inertia (I):

$$I = \frac{\pi r^4}{4} = \frac{\pi(1.0)^4}{4} = 0.785 \text{ mm}^4$$

Critical Buckling Load:

$$P_{cr} = \frac{\pi^2 E_s I}{(Kl)^2}$$

$$P_{cr} = \frac{\pi^2 \times 2500 \times 0.785}{(0.5 \times 14.14)^2}$$

$$P_{cr} = \frac{19368}{49.98}$$

$$P_{cr} = 387.5 \text{ N per strut}$$

Because the axial load required to yield a single 2 mm strut ($F_{yield_strut} = \sigma_{ys} \times \pi r^2 = 40 \times 3.14 = 125.6 \text{ N}$) is significantly lower than the critical buckling load (387.5 N), the analytical model predicts that the structure will fail via plastic yielding of the struts rather than elastic buckling.